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lab4_rc.m

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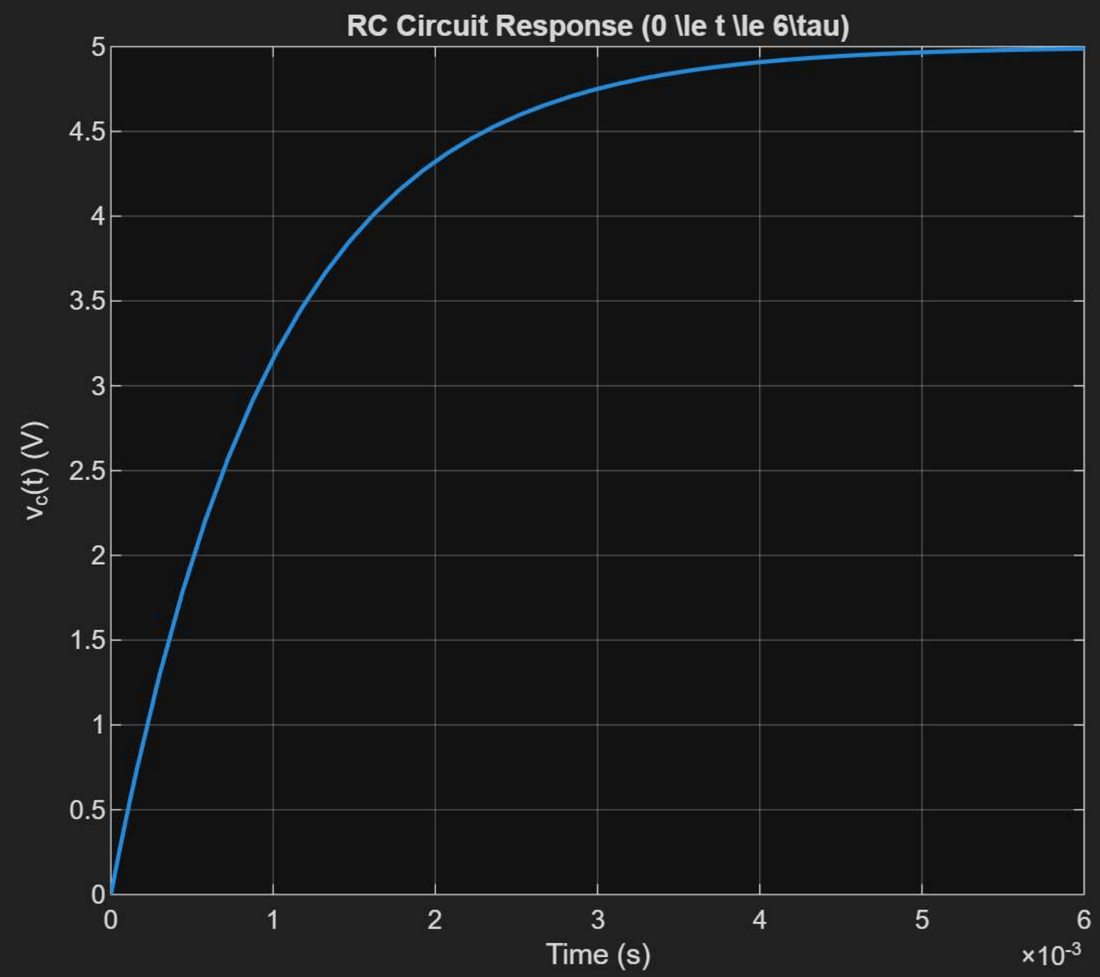
```
clc;clear;close all;

% Lab 4  RC Circuit
% R=1kΩ, C=1μF, vin(t)=5V (t>=0), vc(0)=0          RC * dvc/dt + vc(t) = vir

R = 1e3;      %om
C = 1e-6;    %Farad
Vin = 5;     % eisagogi (Volts)
tau = R*C;    % sthathera Time
tspan = [0 6*tau];
vc0 = 0;
% dvc/dt = (Vin - vc)/(R*C)
f = @(t,vc) (Vin - vc)/(R*C);
[t, vc] = ode45(f, tspan, vc0);
figure;
plot(t, vc, 'LineWidth', 1.6);
grid on;
xlabel('Time (s)');
ylabel('v_c(t) (V)');
title('RC Circuit Response (0 \le t \le 6\tau)');

```

Figure 1



Command Window

```
Warning: Error in state of SceneNode.
String scalar or character vector must have valid interpreter syntax:
RC Circuit Response (0 \\le t \\le 6\\tau)

>> Press Ctrl + Shift + P to generate code with Copilot
```

Editor: 100% UTF-8 CRLF Script Ln 21 Col 1

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lab4_rc.m × lab4_rlc.m ×

/MATLAB Drive/lab4_rlc.m

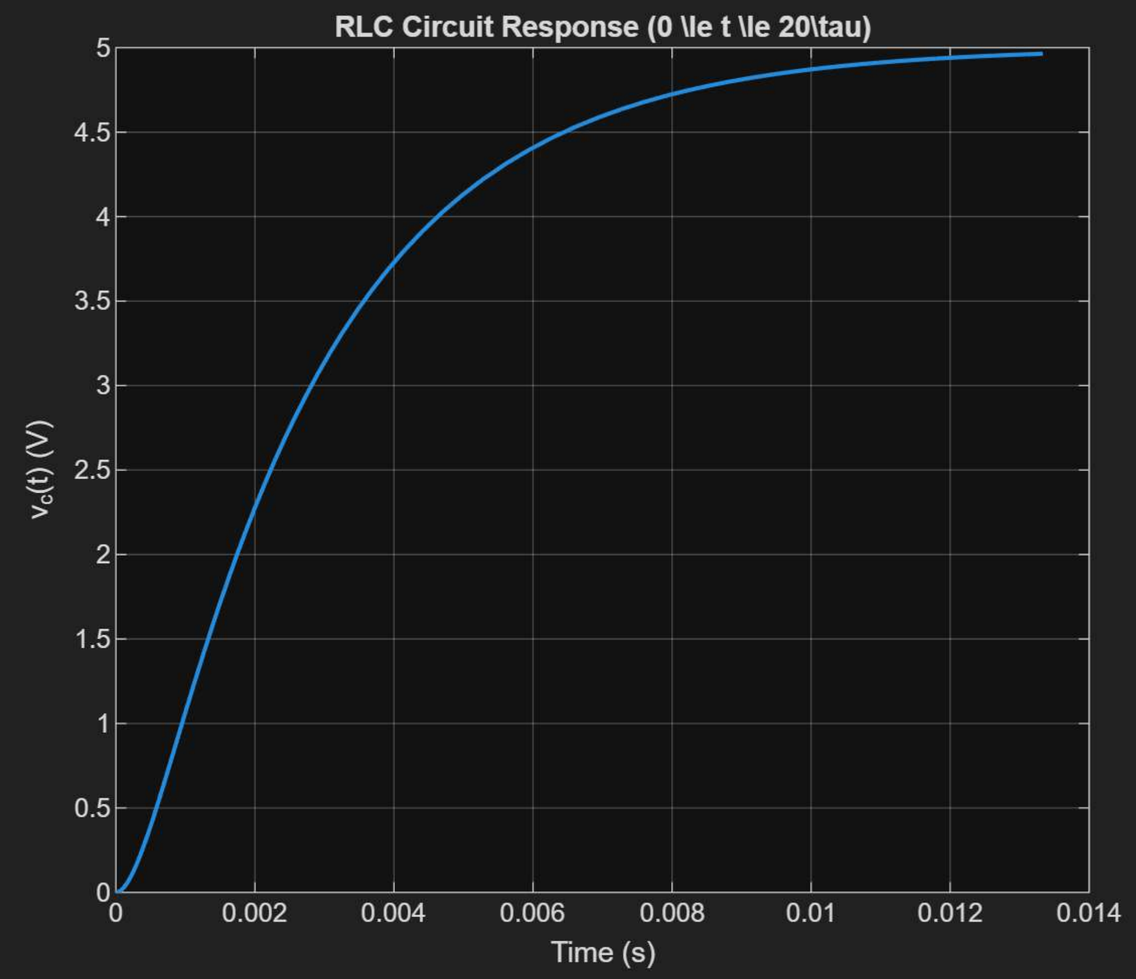
```
clc;clear;close all;

% Lab 4 RLC kikloma
% R=3Ω, C=1mF, L=1mH vc(0)=0, dvc/dt(0)=0

R = 3;        % om
C = 1e-3; %farad
L = 1e-3;     %henry
Vin = 5;    %(Volts)
tau = 2*L/R;  %Time stathera
tspan = [0 20*tau];

%State metablites   x1=vc x2=dvc/dt   x1'=x2    x2'=( (1/C)*Vin - R*x2 - (1/C)*x1 )
f = @(t,x) [ ...
    x(2); ...
    ((Vin/C) - R*x(2) - (1/C)*x(1))/L ...
];
x0 = [0; 0];
[t, x] = ode45(f, tspan, x0);
vc = x(:,1);
figure;
plot(t, vc, 'LineWidth', 1.6);
grid on;
xlabel('Time (s)');
ylabel('v_c(t) (V)');
title('RLC Circuit Response (0 \le t \le 20\tau)');
```

Figure 1 ×



Command Window

```
String scalar or character vector must have valid interpreter syntax:
RLC Circuit Response (0 \\le t \\le 20\\tau)

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```

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